

Research Interests

formal verification of concurrent and distributed systems, with emphasis on weak consistency semantics and compositional reasoning

Education

- 2022–present **PhD CS**, *Northeastern University (NU)*, Boston, MA
4.0/4.0 GPA
- 2020–2022 **MMath Thesis CS**, *University of Waterloo (UW)*, Waterloo, ON
95.75/100.00 GPA
- 2014–2018 **BA CS with Honors & BA Math**, *New York University (NYU)*, New York, NY
3.921/4.000 GPA, Summa Cum Laude, Phi Beta Kappa

Selected Research Experience

- 2022–present **NU PhD Research Assistant**, *Advisor: Ji-Yong Shin*
Developing a top-down verification framework for weakly consistent distributed systems based on novel compositional and operational models. Verified, using this framework, safety and liveness properties of several Go implementations.
- 2020–2022 **UW Master’s Research Assistant**, *Advisor: Martin Karsten*
Thesis: Improving Data Locality in Applications through Execution Delegation
Developed tools and methods for predicting situations where delegating execution between cores via user thread migration can improve an application’s data locality and, therefore, performance.
- 2017–2018 **NYU Senior Honors Thesis**, *Advisor: Christopher Mitchell*
Thesis: Ghosting ASLR: A Spectre Extension
Designed, implemented, and evaluated a novel approach for defeating address space layout randomization by leveraging the Spectre microprocessor vulnerability.
- Summer 2016 **NYU Courant Undergraduate Research Fellow**, *Advisor: Jinyang Li*
Implemented and evaluated a novel database storage structure to reduce the latency of accesses performed in parallel with database entry migration.

Industry Experience

- 2018–2020 **Back-End Research & Development Engineer**, *Geopipe, Inc.*, New York, NY
- Managed and developed the generation of virtual model products.
 - Implemented and tested research algorithms to improve the quality and accuracy of product models. Work funded by NSF SBIR: Phase II grant.
 - Supervised the daily work of undergraduate interns.
- Summer 2017 **Back-End Research & Development Assistant**, *Geopipe, Inc.*, New York, NY
- Spring 2018
- Managed and developed the generation of virtual model products.
 - Assisted in the implementation and testing of research algorithms to improve the quality and accuracy of product models. Work funded by NSF SBIR: Phase I grant.

Selected Awards

- 2025 **SOSP ’25 Student Scholarship & Travel Grant**, ACM

- 2023 **OSDI '23 Student Grant**, *USENIX*
- 2020–2022 **Graduate Excellence Award in CS**, *UW CS*
- 2020 **Graduate Scholarship**, *UW CS*
- 2018 **Best in Panel Presentation: CS, Math**, *NYU CAS 44th Annual Research Conference*
- 2018 **Academic Excellence & Achievement in Science**, *NYU CAS*
- 2018 **CS Prize: Outstanding Performance**, *NYU CAS*
- 2017 **CS Prize: Most Promising Student in the Junior Year**, *NYU CAS*
- 2017 **Dean's Undergraduate Research Fund Award**, *NYU CAS*
- 2014–2017 **Dean's List**, *NYU CAS*

Selected Teaching Experience

- Fall 2024 & Spring 2025 **CS5600 Computer Systems Teaching Assistant**, *NU*
- Spring & Fall 2021 **CS456 Computer Networks Teaching Assistant**, *UW*
- Winter 2021 **CS116 Introduction to CS 2 Teaching Assistant**, *UW*

Selected Presentations

- 2025 **13th Workshop on PLs and OSeS (PLOS 2025)**, Seoul, Korea
Compositional Model-Driven Verification of Weakly Consistent Distributed Systems
- 2024 & 2025 **New England Systems Verification Day**, Boston, USA
- 2025 **New England Systems Day**, Boston, USA
- 2025 **Software Day**, *NU*
- 2024 **Systems Day**, *NU*
- 2022 **Master's Thesis Presentation**, *UW*
Improving Data Locality in Applications through Execution Delegation
- 2018 **44th Undergraduate Research Conference**, *NYU CAS*
Ghosting ASLR: A Spectre Extension

Publications

- Curto, Bryant J.**, Jeonghyeon Kim, Alan Wang, Gijung Im, Jieung Kim, Jeehoon Kang, and Ji-Yong Shin (2025). “Compositional Model-Driven Verification of Weakly Consistent Distributed Systems”. In: *Proceedings of the 13th Workshop on Programming Languages and Operating Systems*. PLOS '25. Seoul, Republic of Korea: Association for Computing Machinery, pp. 42–50. ISBN: 9798400722257. DOI: 10.1145/3764860.3768331. URL: <https://doi.org/10.1145/3764860.3768331>.
- Xiong, Kezhi, Soonwon Moon, Joshua H Kang, **Bryant Curto**, Jieung Kim, and Ji-Yong Shin (2025). “ReCraft: Self-Contained Split, Merge, and Membership Change of Raft Protocol”. In: *2025 55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*. IEEE, pp. 157–170.
- Curto, Bryant J** (2023). “Improving Data Locality in Applications through Execution Delegation”. MA thesis. University of Waterloo.

Patents

Curto, Bryant J, Thomas Daniel Dickerson, and Daniel Christopher Ritchie (Mar. 2023). *Systems and methods for inferring object from aerial imagery*. US Patent App. 17/909,119.

Skills

PLs: Rocq/Coq, C++, C, Python, Bash

OSes: Linux, MacOS

Tools: Vim, Perf, Git, GDB/LLDB, gnuplot, L^AT_EX